QUESTION BANK

S.No	QUESTIONS			
UNIT 1 **Introduction to Deep Learning: Basics: Biological Neuron, Idea of computational units, McCulloch Pitts unit and Thresholding logic, Linear Perceptron, Perceptron Learning Algorithm,Linear separability. Convergence theorem for Perceptron Learning Algorithm**				
1	What is MP neuron?			4M
2	Tell about Thresholding logic.			4M
3	Relate MP neuron with biological neuron.			4M
4	Notes limitations on MP neuron.			4M
5	Explain Perceptron model with architecture?			7M
6	Implement ANDNOT gate using MP Neuron			7M
7	Differentiate between McCulloch-Pits Model and Perceptron model?			7M
8	What are the limitations of theMP neuron model, and how are these limitations addressed by the perceptron model? Discuss in detail.			7M
9	State and prove Convergence theorem for Perceptron Learning Algorithm.			10M
10	What do mean by non-separable dataset. Implement XOR Boolean function using MP neuron.			10M
11	Design model for AND, OR and Tautology function using MP Neuron with geometric interpretation.			10M
12	Design model for NAND, and NOR using MP Neuron and discuss their geometric interpretation			10M
UNIT 2 **Feed forward Networks: Multilayer Perceptron, Gradient Descent, Back propagation, Empirical Risk Minimization.Auto encoders: Regularized Auto encoders, Representational Power, Layer, Size, and Depth ofAuto encoders, Stochastic Encoders and Decoders, Contractive Encoders. Regularization: Bias Variance Tradeoff, L2regularization, Early stopping, Data setaugmentation, Parameter sharing and tying, Injecting noise at input, Ensemble methods,Dropout, Greedy**				
1	Explaingradient descent in ANN?			4M
2	Briefly explain about dropout.			4M
3	What is trainable parameter and hyperparameters?			4M
4	What is the role of the encoder and decoder in an autoencoder architecture?			4M
5	Explain multilayer perceptron with architecture and computations?			7M

6	Tell about (a) Stochastic Autoencoder? (b)Contractive Autoencoder?			7M
7	What is Regularization? Discuss L1, L2 regularization.			7M
8	Differentiate batch, stochastic and minibatch gradient descent.			7M
9	What is Bias, Variance and Trade-off. Explain briefly with example.			10M
10	Explain forward propagation and backpropagation in ANN with example.			10M
11	Discuss following terms (a) Data augmentation (b) Early stopping (c) Vanishing gradient (d) Loss function			10M
12	Explain types of Autoencoders with Architectures			10M
UNIT 3 **Convolutional Neural Networks: The Convolution Operation - Variants of the Basic Convolution Function -Structured Outputs - Data Types – Efficient Convolution Algorithms - Random or Unsupervised Features- LeNet, Alex Net**				
1*	What is feature extraction? Where is it happening in CNN.?			4M
2	What are padding and striding in CNN?			4M
3*	Tell about some basic functionalities of CNN.			4M
4	What do you mean by down sampling? Discuss its advantages in CNN.			4M
5*	What is LeNet. Explain its architecture.			7M
6	What is AlexNet. Explain its architecture.			7M
7*	Compare ANN and CNN with working principles.			7M
8	Calculate the output dimensions (height × width × depth) after each convolutional and pooling layer. Consider that there is at least three convolutional and pooling layers in CNN. Determine total no, of trainable parameter in CNN.			7M
9	Define and explain CNN and operations with neat diagram.			10M
10*	Explain types of CNN Architectures. Briefly			10M
11*	Discuss following terms (a) padding (b) striding (c) fully connected (d) feature map (e) pooling			10M

12	Explain the difference between a convolutional layer and a fully connected layer in a CNN with example.			10M
UNIT 4 **Recurrent Neural Networks: Bidirectional RNNs-Deep Recurrent Networks Recursive Neural Networks- The Long Short- Term Memory and Other Gated RNNs**				
1*	Define and explain RNN with architecture?			4M
2*	What are the types of RNN? Explain with example.			4M
3	What is DEEP RNN? Explain with example			4M
4	What is gradient clipping in RNNs? Explain how it helps in handling the exploding gradient problem with an example.			4M
5*	Define and explain Recursive RNN? with architecture?			7M
6	Explain Gated RNN with architecture?			7M
7*	Differentiate between LSTM and GRU.			7M
8	Write the mathematical equations for the forget, input, and output gates of an LSTM. Briefly explain the function of each gate.			7M
9*	Explain the difference between a standard Recurrent Neural Network (RNN) and a Bidirectional RNN (BiRNN). Give one example of a task where a BiRNN is more suitable than a standard RNN.			10M
10	Explain LSTMRNN with architecture?			10M
11*	Explain the working mechanism of Long Short-Term Memory (LSTM) networks, focusing on the roles of input, forget, and output gates.			10M
12	What is the vanishing gradient problem in standard RNNs? Explain how LSTM and Gated RNNs help to overcome this issue.			10M
UNIT 5 Deep Generative Models: Boltzmann Machines - Restricted Boltzmann Machines - Introduction to MCMC and Gibbs Sampling – gradient computations in RBMs-Deep Belief Networks-Deep Boltzmann Machine				
1*	Define a Boltzmann Machine and explain its use in unsupervised learning.			4M
2*	Differentiate between a Boltzmann Machine and a Restricted Boltzmann Machine.			4M
3	What are the visible layer and hidden			4M

	layers explain?			
4	List three real-world applications of Restricted Boltzmann Machines and explain briefly why RBMs are suitable for these tasks.			4M
5*	Explain DeepBoltzmann machines?			7M
6	Elaborate on the principles of Gibbs Sampling. Discuss its strengths and limitations in the context of generative modelling			7M
7*	Explain about MCMC?			7M
8	Describe the role of **Gibbs Sampling** in RBMs. How is it used to generate new data samples from a trained RBM?			7M
9	Establish the conceptual and architectural connections between Deep Belief Networks (DBNs) and Boltzmann Machines. Explain how the layer-wise training in DBNs is related to the training of Boltzmann Machines.			10M
10*	Discuss the challenges associated with computing gradients in training RBMs.			10M
11	What is the architecture of a Deep Belief Network (DBN)?			10M
12*	Compare Deep Boltzmann Machines (DBM) with Deep Belief Networks (DBN) in terms of architecture, training procedure.			10M